

A Concise Derivation of Principal Component Analysis

Goal

Given data points

$$x_i \in \mathbb{R}^p, \quad i = 1, \dots, n,$$

PCA aims to construct a lower-dimensional representation

$$y_i = U_k^T(x_i - m), \quad y_i \in \mathbb{R}^k, \quad U_k \in \mathbb{R}^{p \times k}, \quad k < p.$$

Here, m is the mean of the data, and the columns of U_k define the k -dimensional subspace onto which the data is projected.

The goal is to choose U_k such that the projection preserves as much information as possible.

In PCA, information is measured by variance. Therefore, PCA chooses the projection directions so that the projected data has maximal variance.

Step 1: Center the Data

The mean of the data is

$$m = \frac{1}{n} \sum_{i=1}^n x_i.$$

Define the centered data points

$$z_i = x_i - m.$$

Centering is necessary because PCA should describe how the data varies around its mean, rather than where the whole dataset is located in space.

Thus, the PCA projection becomes

$$y_i = U_k^T z_i.$$

Step 2: Start with a One-Dimensional Projection

First consider projecting the data onto one direction

$$v \in \mathbb{R}^p.$$

We require

$$\|v\| = 1,$$

or equivalently

$$v^T v = 1.$$

This constraint is necessary because otherwise the projected variance could be made arbitrarily large simply by scaling v .

The projection of z_i onto v is

$$v^T z_i.$$

Therefore, the variance of the projected data is

$$\text{Var}(v) = \frac{1}{n-1} \sum_{i=1}^n (v^T z_i)^2.$$

Step 3: Write the Objective in Matrix Form

Define the centered data matrix

$$Z = [z_1, z_2, \dots, z_n] \in \mathbb{R}^{p \times n}.$$

Define the scatter matrix

$$S = ZZ^T.$$

Then

$$\sum_{i=1}^n (v^T z_i)^2 = v^T ZZ^T v = v^T S v.$$

Hence,

$$\text{Var}(v) = \frac{1}{n-1} v^T S v.$$

Since $\frac{1}{n-1}$ is constant, maximizing the variance is equivalent to solving

$$\max_v \quad v^T S v$$

subject to

$$v^T v = 1.$$

Step 4: Solve the Constrained Optimization Problem

This is a constrained optimization problem, so we use the method of Lagrange multipliers.

Define the Lagrangian

$$\mathcal{L}(v, \lambda) = v^T S v - \lambda(v^T v - 1).$$

To find an optimum, take the derivative with respect to v and set it to zero:

$$\nabla_v \mathcal{L} = 0.$$

Since S is symmetric,

$$\nabla_v (v^T S v) = 2Sv,$$

and

$$\nabla_v (v^T v) = 2v.$$

Therefore,

$$2Sv - 2\lambda v = 0.$$

So

$$Sv = \lambda v.$$

This is an eigenvalue equation. Hence, the optimal projection directions must be eigenvectors of the scatter matrix S .

Step 5: Choose the Direction of Maximal Variance

Suppose v is an eigenvector of S , so

$$Sv = \lambda v.$$

Then

$$v^T Sv = v^T (\lambda v) = \lambda v^T v.$$

Because

$$v^T v = 1,$$

we get

$$v^T Sv = \lambda.$$

Therefore, the variance along direction v is proportional to the eigenvalue λ .

Thus, to maximize the projected variance, we choose the eigenvector with the largest eigenvalue. This direction is called the first principal component.

Step 6: Extend to k Dimensions

For a k -dimensional projection, we choose k orthonormal directions

$$u_1, u_2, \dots, u_k.$$

Collect them into the matrix

$$U_k = [u_1, u_2, \dots, u_k] \in \mathbb{R}^{p \times k}.$$

The orthonormality condition is

$$U_k^T U_k = I_k.$$

The projected representation is

$$y_i = U_k^T z_i = U_k^T (x_i - m).$$

Since each eigenvalue measures the variance preserved by its eigenvector, the best k -dimensional subspace is obtained by choosing the eigenvectors corresponding to the k largest eigenvalues:

$$U_k = [u_1, u_2, \dots, u_k],$$

where

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p.$$

Step 7: Reconstruction and Approximation Error

The projected coordinates are

$$y_i = U_k^T z_i.$$

The reconstruction in the original space is

$$\hat{z}_i = U_k y_i.$$

Substituting $y_i = U_k^T z_i$, we get

$$\hat{z}_i = U_k U_k^T z_i.$$

Thus, the reconstruction of the original point is

$$\hat{x}_i = m + U_k U_k^T (x_i - m).$$

The reconstruction error is

$$z_i - \hat{z}_i = z_i - U_k U_k^T z_i = (I - U_k U_k^T) z_i.$$

Therefore, the total approximation error is

$$\sum_{i=1}^n \|(I - U_k U_k^T)(x_i - m)\|^2.$$

Choosing the eigenvectors with the largest eigenvalues maximizes the retained variance. Since the total variance is fixed, maximizing retained variance is equivalent to minimizing the remaining reconstruction error.

Conclusion

PCA starts from the goal

$$y_i = U_k^T (x_i - m),$$

where y_i is a lower-dimensional representation of x_i .

To preserve as much information as possible, PCA maximizes the variance of the projected data. This leads to the constrained optimization problem

$$\max_v v^T S v \quad \text{subject to} \quad v^T v = 1.$$

Using Lagrange multipliers, this becomes the eigenvalue problem

$$Sv = \lambda v.$$

Therefore, PCA chooses the eigenvectors of the scatter matrix with the largest eigenvalues. These directions preserve the largest amount of variance and minimize the approximation error.

PCA reduces dimensionality by projecting data onto the directions of maximal variance.